相位匹配理论

1. 相位匹配理论.

2. 基于单轴晶体双折射相位匹配

3. 双折射相位匹配中走离效应.

相位匹配的意义 —— 二次谐波.

二次谐波耦合方程：

$$\begin{cases} \dfrac{dE_1(z)}{dz} = \dfrac{2i d_{eff}\omega_1^2}{k_1 c^2} E_2(z) E_1^*(z) e^{-i\Delta k z} \\ \dfrac{dE_2(z)}{dz} = \dfrac{i d_{eff}\omega_2^2}{k_2 c^2} E_1^2(z) e^{i\Delta k z} \end{cases}$$

其中，$\omega_2 = 2\omega_1$.

$$\Delta k = 2k_1 - k_2 = 2\frac{2\pi n(\omega_1)}{\lambda_1} - \frac{2\pi n(\omega_2)}{\lambda_2} = 2\frac{\omega_1 n(\omega_1)}{c} - \frac{\omega_2 n(\omega_2)}{c}$$

作基波小信号近似：

$$E_2(z) = \frac{i d_{eff}\omega_2^2}{k_2 c^2} E_1^2 \cdot L\, e^{i\Delta k L/2} \operatorname{sinc}\frac{\Delta k L}{2}$$

$$= \frac{i d_{eff}\omega_2^2}{k_2 c^2} E_1^2 \cdot \frac{2}{\Delta k} e^{i\Delta k L/2} \operatorname{sinc}\frac{\Delta k L}{2}$$

又 $I_i = \frac{1}{2} n_i \varepsilon_0 c |E_i|^2 = \frac{P}{S}$，$P_{peak} = \frac{\langle P\rangle}{\Delta t \cdot f}$，

$$\langle P_2(L)\rangle = \frac{8 d_{eff}^2 \omega_2^2}{\varepsilon_0 n_1^2 n_2 c^3} \frac{\langle P_1\rangle^2}{S\Delta t f} \frac{\sin^2\frac{\Delta k L}{2}}{\Delta k^2}$$

$P_2 \sim 10^{-12}$ (1W，连续光)

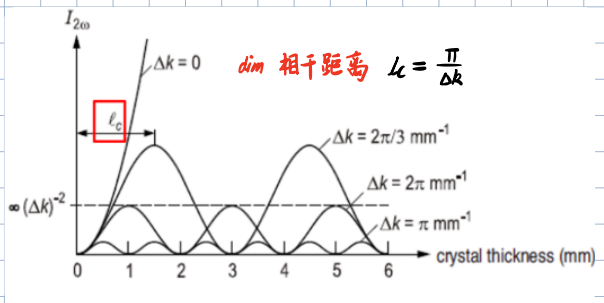
when $\Delta k = 2k_1 - k_2 = 0$，基波小信号近似基波不变，

$$E_2(z) = \frac{i d_{eff}\omega_2^2}{k_2 c^2} E_1^2 \cdot L$$

$$\langle P_2(L)\rangle = \frac{2 d_{eff}^2 \omega_2^2 L^2}{\varepsilon_0 n_1^2 n_2 c^3} \frac{\langle P_1\rangle^2}{S\Delta t f}$$

$\langle P_2(L) \rangle \approx L^2$. $(\Delta k = 0)$

$\langle P_2(L) \rangle \approx \sin^2 \frac{\Delta k L}{2}$ $\Delta k \neq 0$.



$P_2 \sim 10^{-6}$ (1W, 连续光 $L = 1$ mm)

相位匹配定义 Phase matching

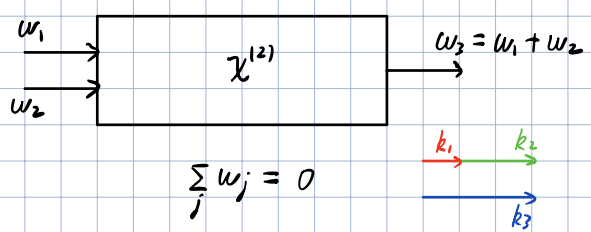
$$\begin{cases} \Delta k = 2k_1 - k_2 = 0 & \text{二次谐波.} \\ \Delta k = k_1 + k_2 - k_3 = 0 & \text{和频过程.} \end{cases}$$

一般形式，n个光场相互作用.(正频率 $k_j > 0$，负频率 $k_j < 0$.)

$$\Delta k = \sum_j \vec{k}_j = 0.$$

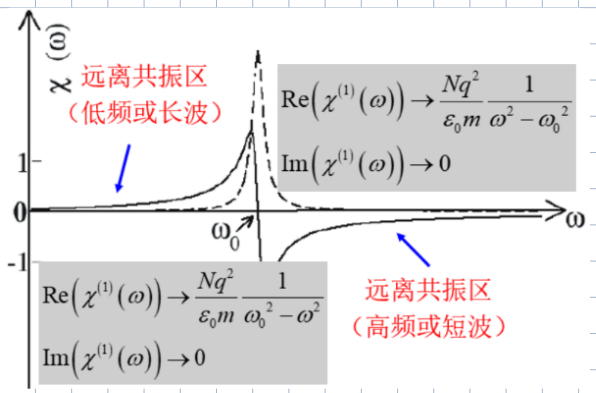
狭义的相位匹配：利用晶体双折射效应弥补材料色散，使波矢失配量为0.

广义的～：通过某种方法实现之.



$$\sum_j \omega_j = 0$$

色散引起相位失配 —— 倍频（or 二次谐波）



振幅：

$$\chi^{(2)}(\omega_3;\omega_1,\omega_2)=\frac{aN^3g}{\varepsilon_0 m^2}\frac{1}{D(\omega_3)D(\omega_1)D(\omega_2)}$$

$$=\frac{a\varepsilon_0^2 m^1}{N^2 g^3}\chi^{(1)}(\omega_3)\chi^{(1)}(\omega_1)\chi^{(1)}(\omega_2)$$

相位：

$$\Delta k=2k_1-k_2=\frac{\omega_2}{c}[n_2(\omega_2)-n_1(\omega_1)]=0$$

注意：

1. 克莱门对称性要求光场处在无色散区。这是在长波段的近似条件，因为非线性极化系数主要影响光场的强度。在长波段，非线性系数主要依赖于共振频率。

2. 对材料介电常数，由于色散会引起光场的相位、而光场相位变化非常敏感，因此，即使在长波段，介电常数的色散不能忽略。

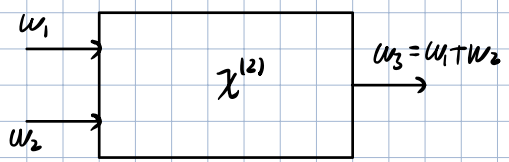
3. ∵介电常数存在色散，需特殊技术手段实现 $\Delta k=0$。

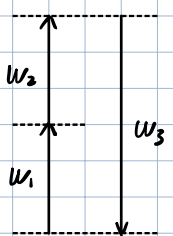
光子角度理解完美相位匹配

频率守恒：$\sum_j \omega_j = 0$ ⟶ 光子能量守恒 $\hbar\sum_j \omega_j = 0$

波矢守恒：$\sum_j \vec{k}_j = 0$ 光子动量守恒 $\hbar\sum_j \vec{k}_j = 0$

三波耦合

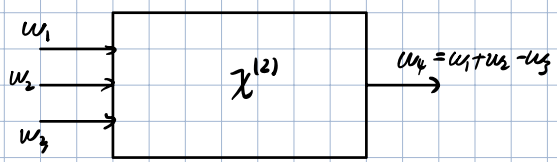


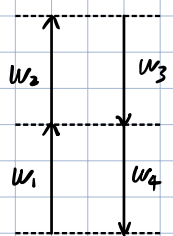


$\hbar\omega_1 + \hbar\omega_2 - \hbar\omega_3 = 0$

$\hbar\vec{k}_1 + \hbar\vec{k}_2 - \hbar\vec{k}_3 = 0$

四波耦合.





$\hbar\omega_1 + \hbar\omega_2 - \hbar\omega_3 - \hbar\omega_4 = 0$

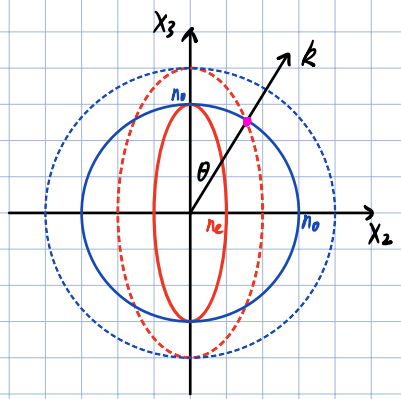
$\hbar\vec{k}_1 + \hbar\vec{k}_2 - \hbar\vec{k}_3 - \hbar\vec{k}_4 = 0$.

微观机制：构成材料系统所有电耦极子保持适当相位，使其辐射的非线性光场相干增强.

可利用晶体双折射使 $\Delta k = 2k_1 - k_2 = 0$ 弥补色散.

一类相位匹配：

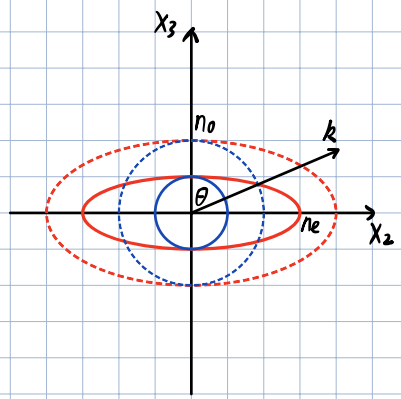
倍频 相位匹配条件：



负单轴晶体

$o + o \to e$

$n_e(2\omega, T, \theta) = n_o(\omega, T)$.



正单轴晶体.

$e + e \to o$

$n_e(\omega, T, \theta) = n_o(2\omega, T)$.

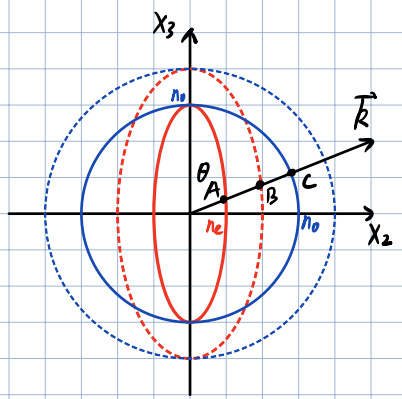
$n_e(\omega, T, \theta)$: ——

$n_o(\omega, T)$: ——

$n_e(2\omega, T, \theta)$: - - -

$n_o(2\omega, T)$: - - -

二类相位匹配条件：

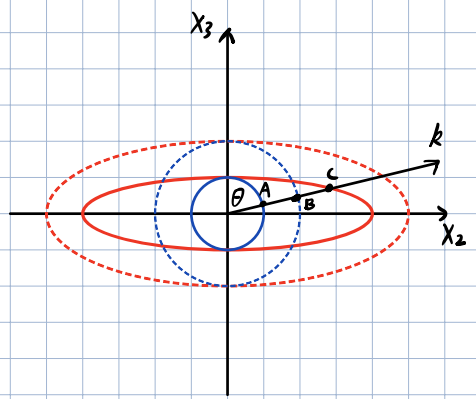


负单轴晶体

$o + e \rightarrow e$

$\frac{1}{2}[n_o(\omega, T) + n_e(\omega, T, \theta)] = n_e(2\omega, T, \theta)$

$(|AB| = |BC|)$



正单轴晶体

$o + e \rightarrow o$

$\frac{1}{2}[n_o(\omega, T) + n_e(\omega, T, \theta)] = n_o(2\omega, T)$

$(|AB| = |BC|)$.

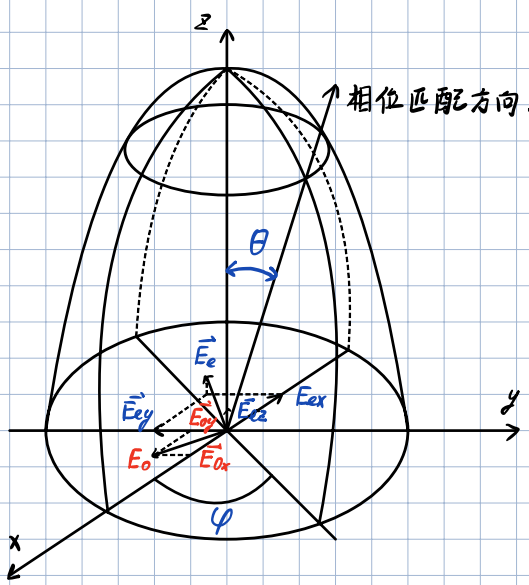
说明：

1. 双折射能够弥补色散的材料才能实现相位匹配，其中，基波利用较大折射率，倍频利用较小折射率.
2. 所有相位匹配角 θ_m 与原点构成1锥面，即存在无数匹配方向.
3. 相位匹配角仅受 θ 限制，通过改变 φ，可对非线性系数进行选择.
4. 存在第二类匹配，一定存在一类匹配；存在一类匹配未必存在二类匹配.

有效非线性系数 $F(\theta, \varphi, \chi)$（即 $2d_{eff}$）计算.

$$\vec{E}_o(\omega) = \vec{E}_o(\omega)\begin{pmatrix}\sin\varphi\\-\cos\varphi\\0\end{pmatrix}, \quad \vec{E}_e(\omega) = E_e(\omega)\begin{pmatrix}-\cos\theta\cos\varphi\\-\cos\theta\sin\varphi\\\sin\theta\end{pmatrix}$$

$$\vec{P}(2\omega) = \varepsilon_0 \overleftrightarrow{\chi} \vec{E}_m(\omega)\vec{E}_n(\omega).$$



- m、n 分别取 o 或 e. 代表 o 光或 e 光单独或相互作用产生的非线性极化
- 相位匹配不受 φ 角限制.

$$\begin{pmatrix} P_1(2\omega) \\ P_2(2\omega) \\ P_3(2\omega) \end{pmatrix} = \varepsilon_0 \overleftrightarrow{\chi} \begin{pmatrix} E_{m,1}(\omega)\, E_{n,1}(\omega) \\ E_{m,2}(\omega)\, E_{n,2}(\omega) \\ E_{m,3}(\omega)\, E_{n,3}(\omega) \\ E_{m,2}(\omega) E_{n,3}(\omega) + E_{n,2}(\omega) E_{m,3}(\omega) \\ E_{m,1}(\omega) E_{n,3}(\omega) + E_{n,3}(\omega) E_{m,3}(\omega) \\ E_{m,1}(\omega) E_{n,2}(\omega) + E_{n,1}(\omega) E_{m,2}(\omega) \end{pmatrix}$$

$$\begin{pmatrix} E_{o1} \\ E_{o2} \\ E_{o3} \end{pmatrix} = E_o(\omega) \begin{pmatrix} \sin\varphi \\ -\cos\varphi \\ 0 \end{pmatrix}, \quad \begin{pmatrix} E_{e1} \\ E_{e2} \\ E_{e3} \end{pmatrix} = E_e(\omega) \begin{pmatrix} -\cos\theta\cos\varphi \\ -\cos\theta\sin\varphi \\ \sin\theta \end{pmatrix}$$

$$\begin{pmatrix} P_1(2\omega) \\ P_2(2\omega) \\ P_3(2\omega) \end{pmatrix} \xrightarrow{o+o\to e} = \varepsilon_0 E_o^2(\omega) \overleftrightarrow{\chi} \begin{pmatrix} \sin^2\varphi \\ \cos^2\varphi \\ 0 \\ 0 \\ 0 \\ -\sin 2\varphi \end{pmatrix} \begin{pmatrix} P_1(2\omega) \\ P_2(2\omega) \\ P_3(2\omega) \end{pmatrix}$$

$$\xrightarrow{e+e\to o} = \varepsilon_0 E_e^2(\omega) \overleftrightarrow{\chi} \begin{pmatrix} \cos^2\theta\cos^2\varphi \\ \cos\theta\sin^2\varphi \\ \sin^2\theta \\ -\sin 2\theta\sin\varphi \\ -\sin 2\theta\cos\varphi \\ \cos^2\theta\sin 2\varphi \end{pmatrix} \begin{pmatrix} P_1(2\omega) \\ P_2(2\omega) \\ P_3(2\omega) \end{pmatrix}$$

$o+e \to o$ 或 $o+e \to e$

$$= \varepsilon_0 E_o(\omega) E_e(\omega) \vec{\chi} \begin{pmatrix} -\frac{1}{2}\cos\theta \sin 2\varphi \\ \frac{1}{2}\cos\theta \sin 2\varphi \\ 0 \\ -\sin\theta \cos\varphi \\ \sin\theta \sin\varphi \\ \cos\theta \cos 2\varphi \end{pmatrix}$$

- m、n 分别取 e 或 o，四种组合均能产生非线性效应. 但受相位匹配条件约束，只有一种可高效产生二次谐波.